

# Bayes Update Cards Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

|            |             |      |                    |
|------------|-------------|------|--------------------|
| Category   | Mathematics | Time | 35-45 min          |
| Difficulty | Medium      | ID   | bayes-update-cards |

Big idea: New evidence updates prior probability; it does not replace the background rate.

## Overview

Students use natural-frequency cards to see how prior probability and new evidence combine in Bayesian updating.

## Student Prompt

Inspect Bayes Update Cards Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

## Concrete Stimulus Packet

|                                   |                                                                                                                                                                                                                                                   |                                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Bayes Update Cards Stimulus A     | Student-facing stimulus 1: Fictional case: 10 of 1000 people have Condition X; a positive test catches 9 real cases and falsely flags 99 non-cases. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof. | Students inspect this exact card first and commit to a claim before hearing anyone else.                                                        |
| Bayes Update Cards Stimulus B     | Student-facing stimulus 2: Notation contrast card: $P(H   +)$ is not the same as $P(+   H)$ . Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.                                                       | Use this for comparison, a second round, or a partner challenge.                                                                                |
| Bayes Update Cards Reveal / Twist | Student-facing stimulus 3: Notation contrast card: $P(H   +)$ is not the same as $P(+   H)$ . Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.                                                       | Teacher reveal: connect the changed response to the big idea — New evidence updates prior probability; it does not replace the background rate. |

## Actual Lesson Questions and Key

| # | Question for students                                                               | Teacher key / cue                                                                                             |
|---|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1 | After inspecting Bayes Update Cards Stimulus A, what is your first knowledge claim? | Teacher cue: look for a clear claim, not just a description.                                                  |
| 2 | Which exact detail from Bayes Update Cards Stimulus A did you use as evidence?      | Teacher cue: students should name visible words, data, source features, method features, or contextual clues. |
| 3 | What did you infer beyond the evidence?                                             | Teacher cue: this is where assumptions, framing, models, or perspective usually appear.                       |
| 4 | How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?                            | Teacher cue: confidence is data for the debrief, not a grade.                                                 |
| 5 | What missing information would most improve the knowledge claim?                    | Teacher cue: push for method, source, context, comparison, or definitions.                                    |
| 6 | How does notation shape mathematical knowledge?                                     | Teacher cue: students should move from the classroom example to a general TOK issue.                          |

## Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how new evidence updates prior probability; it does not replace the background rate.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

## Setup Checklist

- Prepare counters in two colours: true positives and false positives.
- Hide notation at first; reveal  $P(H | +)$  only after students build the frequency table.

## Ready-to-Use Examples

- Fictional case: 10 of 1000 people have Condition X; a positive test catches 9 real cases and falsely flags 99 non-cases.
- Notation contrast card:  $P(H | +)$  is not the same as  $P(+ | H)$ .

## Suggested Timing

| Stage   | Teacher move                                                                              | Watch for                                                                   |
|---------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1       | Give students a fictional condition, base rate, and test accuracy in words.               | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 2       | Ask for an intuitive estimate after a positive result.                                    | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 3       | Build the case using natural frequencies, separating true positives from false positives. | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 4       | Introduce the update calculation $P(H   +) = 9 / (9 + 99) = 1 / 12 \approx 8.3\%$ .       | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 5       | Compare how the same information feels in words, percentages, frequencies, and notation.  | Assumptions, confidence shifts, evidence claims, and language choices.      |
| Debrief | Move from the activity result to a TOK knowledge question.                                | Students distinguishing method, evidence, interpretation, and overclaiming. |

## Teacher Language

- The notation is not decoration; it tells us which direction the probability is asking about.
- Point to the numerator and denominator before you calculate.

## Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

## Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

## Differentiation and Adaptations

- Support: use 100 people and round numbers.
- Challenge: change the base rate and ask students to predict the direction of the update before calculating.

## **Extension Tasks**

- Students write a public-health risk statement in three formats and critique each one.
- Compare Bayesian updating with search rankings, eyewitness evidence, or historical source reliability.

## **Related Activities**

- Base Rate Neglect Clinic
- Monty Hall Shock
- Calculator Trust Test